

PROGRAM – 06: Deploy an automated build pipeline using Docker hub

Structure of the program:

- Dockerfile
- Package.json
- Package-lock.json
- Node_modules

Steps:

1. Create the personal access token in github

- Create a repository(i.e program-6)
- Navigate to the settings -> Developers settings -> Generate tokens
- Generate the token with specific scope:
 - Write:packages
 - Read:packages

2. Create a Docker file and build the images

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ nano Dockerfile
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ cat Dockerfile
FROM python:3.9-slim

WORKDIR app/

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000

CMD ["flask", "run"]
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ █
```

```

1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ ls
app.py Dockerfile requirements.txt
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ sudo docker build -t program-6 .
[sudo] password for 1rv24mc017_ananya_ganapati_hegde:
[+] Building 9.7s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 255B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
=> [internal] load build context
=> => transferring context: 516B
=> CACHED [2/5] WORKDIR app/
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:1a1bd693803359adbfe479558a514730a42824a9281ec065da3bb3aa8a42f39
=> => naming to docker.io/library/program-6

1 warning found (use docker --debug to expand):
- WorkdirRelativePath: Relative workdir "app/" can have unexpected results if the base image changes (line 3)
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$

```

3. Tag the image for docker hub and login to the docker hub using PAT

```

1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ docker tag program-6 unique1112/program-6:latest
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ docker login -u unique1112

```

i Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit <https://app.docker.com/settings>

Password:
Login Succeeded

4. Push the image to the docker hub

```

1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ sudo docker push unique1112/program-6:latest
The push refers to repository [docker.io/unique1112/program-6]
c113fd1bfbd1: Mounted from unique1112/program6
c783fc9c2b3b: Mounted from unique1112/program6
6c23d015d991: Mounted from unique1112/program6
f9ea446119e4: Pushed
c8f6b54339a8: Mounted from unique1112/program6
298992e09a03: Mounted from unique1112/program6
4f237755fbae: Mounted from unique1112/program6
d7c97cb6f1fe: Mounted from unique1112/program6
latest: digest: sha256:df12a1cc6a0cc8262dd847ea9bc61aebcbaa4187c3108b8631e41e47c19c989c size: 1990
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$

```

5. login to the github container registry using PAT

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ sudo docker login ghcr.io -u ananyahegde
Password:
Login Succeeded
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$
```

6. Tag the image for the github and Push the image to the github

```
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ sudo docker tag program-6 ghcr.io/ananyahegde/program-6:latest
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ sudo docker push ghcr.io/ananyahegde/program-6:latest
The push refers to repository [ghcr.io/ananyahegde/program-6]
c113fd1bfbd1: Mounted from ananyahegde/program6
c783fc9c2b3b: Mounted from ananyahegde/program6
6c23d015d991: Mounted from ananyahegde/program6
f9ea446119e4: Mounted from ananyahegde/program6
c8f6b54339a8: Mounted from ananyahegde/program6
298992e09a03: Mounted from ananyahegde/program6
4f237755fbae: Mounted from ananyahegde/program6
d7c97cb6f1fe: Mounted from ananyahegde/program6
latest: digest: sha256:b8e6d715a143e3eeb494a6cebf5d79568c00bc7ac4b2d00c7d073ac760cdefd1 size: 1990
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ █
```

7. Add the local project to the github:

- Add the remote repository(if not already linked)

git remote add origin <https://github.com/<username>/program-6.git>

- If the remote repository already exists(but incorrect),

git remote set-url origin <https://github.com/<username>/program-6.git>

8. Add the source code to the remote repository

```

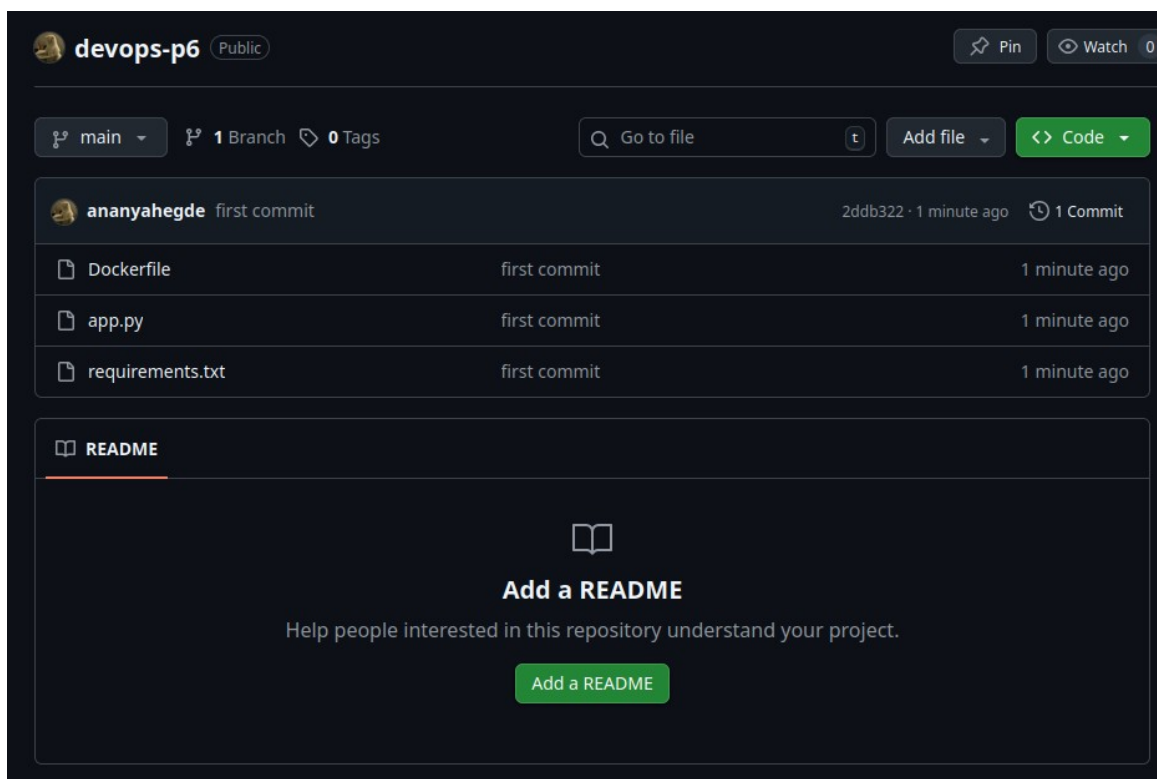
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/1rv24mc017_ananya_ganapati_hegde/DevOps/lab/p6/.git/
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ git add .
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ git commit -m "first commit"
[master (root-commit) 2ddb322] first commit
 3 files changed, 25 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 app.py
 create mode 100644 requirements.txt
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$ git status
On branch master
nothing to commit, working tree clean
1rv24mc017_ananya_ganapati_hegde@ayla-victus:~/DevOps/lab/p6$

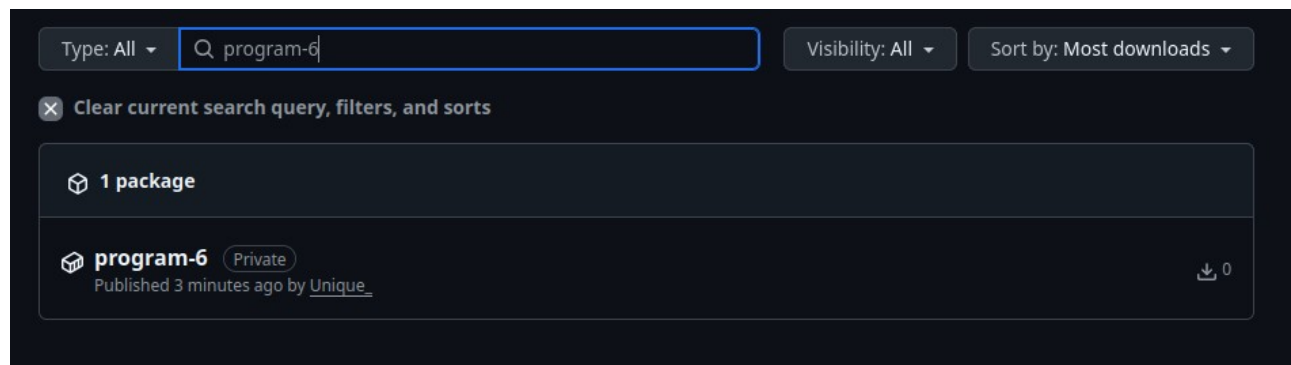
```

9. Check the uploads in the both registries

In the github:

- Check whether the source code is present in the repository
- Under your profile in the packages the particular image is present.





In the dockerhub:

- Under your repository check whether the particular image

